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FUTURES LAB
FOR DEMOCRACY, TECH & HUMAN RIGHTS

NEUROTECHNOLOGY AND THE ADMINISTRATION OF JUSTICE

Submission to the Office of the UN High Commissioner for Human Rights for Input to the Report of the UN Secretary-General Pursuant to UN General Assembly Resolution A/RES/79/172

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CONTENTS

1. Executive Summary 1
2. Latest Developments in Neurotechnology and Justice 2
3. Regulatory Framework 3
4. Data Protection: Law and Enforcement 4
5. Human Rights Risks and Needed Safeguards 4
6. Human Rights Protections 5
7. Human Rights Impacts 6
8. Impacts on Minority Groups 6
9. Accountability and Access to Remedies 7
10. Judicial Decisions 8

1. Executive Summary

The rapid advancement of neurotechnology, particularly brain-computer interfaces (BCIs), presents transformative opportunities alongside human rights challenges. In 2024, the United States' BRAIN Initiative achieved significant breakthroughs, including high-accuracy decoding of internal speech and advanced brain mapping. Private sector innovation has accelerated in parallel. Companies such as Neuralink and China's NeuroXess are advancing implantable BCIs that translate neural signals into language and movement, while new chip technologies promise scalable commercialization. Although these developments are framed as life-changing tools for individuals with paralysis or neurological conditions, they remain in

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clinical stages and lack clear pricing structures, raising concerns about accessibility, equity, and long-term oversight.

Governance frameworks have not kept pace with innovation. Regulatory authority is fragmented across jurisdictions, and decision-making processes often privilege scientific, corporate, and state actors while marginalizing those most affected: patients, defendants, prisoners, and vulnerable communities. Uneven participation reinforces structural inequality. Human Rights Impact Assessments (HRIAs) are increasingly recognized as essential ex-ante safeguards, ensuring risks to privacy, autonomy, and mental integrity are evaluated before deployment. The case of *Girardi v. Emotiv Inc.*, in which Chile's supreme court ordered the deletion of neural data and suspended sales pending public health certification, demonstrates that commercialization cannot rely solely on click-through consent.

Legal protections vary widely. Within the United States, states such as California and Colorado have amended privacy laws to include neural data, yet no federal standard exists. By contrast, the European Union's General Data Protection Regulation and Artificial Intelligence Act embed human dignity and data protection into binding supranational law. Globally, there remains no unified international framework for neurorights, despite early leadership from countries such as Chile and guidance from organizations such as UNESCO and the Organization for Economic Cooperation and Development.

Emerging neurotechnologies also amplify risks of discrimination. Biased datasets and predictive systems may disproportionately harm people of African descent, queer individuals, and other marginalized groups, particularly in policing, employment, and judicial contexts. The potential for cognitive surveillance, involuntary inference of identity traits, and unequal access to benefits underscores the urgency of enforceable safeguards.

Given these developments, the United Nations should establish global standards grounded in human dignity, cognitive liberty, and equality. Proactive oversight, independent audits, clear consent limitations, and meaningful participation of impacted communities are critical to ensuring neurotechnology advances justice rather than undermining fundamental human rights.

2. Latest Developments in Neurotechnology and Justice

In 2024, the United States' BRAIN initiative made notable developments in various fields, using BCI (Brain Computer Interface) to translate internal speech with 79 percent accuracy onto a monitor, publish the highest-to-date 3D resolution map of the human brain tissue, and workshop the Clinical Outcome Assessments with the Food and Drug Administration.[2] But in 2025, the BRAIN initiative budget dropped by $81 million from the previous year due to congressional decreases of the 21st Cures Act funding, creating a slight worry for the future of BRAIN after their FY2026 budget expires.[3]

Other notable advances include Elon Musk's Neuralink: Telepathy, which allows the user to control a computer interface with their thoughts. Currently, it maintains 21 clinical patients worldwide.[4]

To rival Neuralink, the Chinese NeuroXess has made a breakthrough as the world's first to simultaneously decode language and movement in real time in a 21-year-old patient with

[2] The BRAIN Initiative Alliance. "A Look Back on the BRAIN Initiative in 2024 (and What's Coming in 2025!) - the BRAIN Initiative Alliance," March 7, 2025.
[3] "Understanding the BRAIN Initiative Budget | BRAIN Initiative," n.d., https://braininitiative.nih.gov/funding/understanding-brain-initiative-budget.
[4] Neuralink. "Two Years of Telepathy | Updates | Neuralink." Neuralink Updates, February 2, 2026.

epilepsy.[5] Additionally, there are reports of a new silicone chip, much smaller and more efficient for brain computer interface, under development by Kampto Neurotech in the United States for commercial versions before preclinical research applications.[6]

All these neurotechnological improvements are advertised as aids to people with severe physical and neurologic disabilities. However, there are major concerns surrounding the accessibility of these services and the lack of legal regulations.

3. Regulatory Framework

In governance of neurotechnology in the administration of justice, decision-making powers are fragmented. Regulatory discussions involve diverse stakeholders, including scientists, businesses, human rights experts, and policymakers who often have different vocabularies, worldviews, and standards of success.[7] The lack of explicit consensus across regulation strategies and disparate regulation recommendations have led to confusion and uneven compliance.[8] This process fails to meaningfully engage citizens, marginalizing those most impacted. Failure of inclusion violates human rights and procedural justice while entrenching institutional and private interests, risking neurotechnology advancing as a product for market expansion while disregarding the human costs.

France's Civil Code exemplifies this paradigm. France was the first country to allow brain imaging in judicial proceedings with consent, for medical or scientific research, or for forensic expertise, except for functional brain imaging.[9] This policy was adopted prior to consensus among neuroscience experts and in opposition to the recommendations of the French Parliamentary Office for the Evaluation of Scientific and Technological Choices. Critics argue there is a lack of distinction among imaging types and unclear limits on use. These critiques illustrate the uneven participation and lack of consultation in a top-down, non-consultative lawmaking process.

Those directly impacted by neurotechnology, including defendants and prisoners, patients and users, vulnerable groups, and marginalized communities, are regarded as secondary advisors or "objects" in the regulatory process. At the global level, perspectives from low- and middle-income countries and the Global South are underrepresented.[10] The human right to citizen participation, with respect to those most directly impacted, requires that citizens be given decision-making powers. The demand for independent audits and evaluations, establishment of redlines, and binding recommendations to enforcement agencies is critical to prevent regulatory capture. While there is promise in neurotechnology's assistance in assessing guilt, accountability, and the risk of recidivism in judicial proceedings, it also carries risks regarding access to the regulatory process.

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[5] 董森河. "BMI Technology Makes Breakthrough in Simultaneously Decoding Language and Movement." Chinadaily.com.cn, n.d.

[6] Jung, Taesung, Nanyu Zeng, Jason D. Fabbri, Guy Eichler, Zhe Li, Erfan Zabeh, Anup Das, et al. "A Wireless Subdural-contained Brain–computer Interface With 65,536 Electrodes and 1,024 Channels." *Nature Electronics* 8, no. 12 (December 8, 2025): 1272–88.

[7] Erica Harper, *The Evolving Neurotechnology Landscape: Examining the Role and Importance of Human Rights in Regulation* (Geneva: Geneva Academy of International Humanitarian Law and Human Rights, Research Brief, December 2023).

[8] Liam J. Robertson, Nathan L. Higgins, Moritz J. Maier, Adrian Carter, and John G. Gardner, "Ethical Governance Strategies for the Responsible Innovation of Neurotechnologies: A Scoping Review," *Journal of Bioethical Inquiry* 22, no. 4 (2025): 931–948.

[9] France, Code Civil (1804), art. 16-14.

[10] Liam J. Robertson, Nathan L. Higgins, Moritz J. Maier, Adrian Carter, and John G. Gardner, "Ethical Governance Strategies for the Responsible Innovation of Neurotechnologies: A Scoping Review," *Journal of Bioethical Inquiry* 22, no. 4 (2025): 931–948.

4. Data Protection: Law and Enforcement

In the United States, some states have worked towards creating protections for neural privacy. California and Colorado are currently the only states to have passed legislation on neural data privacy, while a dozen others have legislation pending.

The California law grants consumers privacy rights regarding personal information and neural data. In this law, neural data is defined as "information that is generated by measuring the activity of a consumer's central or peripheral nervous system, and that is not inferred from nonneural information."[11] The Colorado Privacy Act regulates "biological, genetic, biochemical, physiological, or neural properties, compositions, or activities or of an individual's body or bodily function" as sensitive data that is to be protected from business use.[12] California and Colorado demonstrate that there is no national standard on which biological data qualifies for privacy protections.

The European Union offers protection for the processing of neural and biometric data through guarantees of fundamental rights, data-protection laws, and criminal-procedure safeguards. The General Data Protection Regulation (GDPR), established in 2018, ensures that personal and sensitive neurobiometric data about health and mental status are legally classified and protected.[13] Only under specifications such as legal necessity, proportionality, and security may courts request the processing of said data.[14] However, there is no international standard for the protection of neural and biometric data, and governments are insufficiently regulating new and emerging technologies. While general data privacy frameworks may provide some protections, specific international standards must be established to recognize the highly sensitive and personal nature of bio- and neurotechnologies.

5. Human Rights Risks and Needed Safeguards

In the implementation of neurotechnology in the administration of justice, there is a critical need to assess risks and implement safeguards. Although UNESCO's Universal Declaration on Bioethics and Human Rights in Article 6 mandates that "all scientific and medical interventions must be carried out with the individual's prior, free, and informed consent," neurotechnology has the capacity to violate personal sovereignty over one's brain in ways that challenge traditional notions of autonomy, privacy, and personal identity.[15]

Neurorights safeguard fundamental aspects of human dignity, ensuring that mental processes and cognitive functions remain inviolable. They align with Articles 12 and 18 of the Universal Declaration of Human Rights (UDHR) which protect privacy and freedom of thought, conscience, and religion.[16]

a. While the first international neurotechnological standard was set in 2019 by the Council of the Organization for Economic Cooperation and Development (OECD), **stricter legal protections must be enacted to protect the human right to sovereignty over one's brain;**

[11] California State Legislature. Senate Bill 1223: Consumer Privacy: Sensitive Personal Information: Neural Data. Chapter 887, Statutes of 2024. Approved September 28, 2024.
[12] Colorado General Assembly. House Bill 24-1058: Protect Privacy of Biological Data. 2024 Regular Session. Approved April 17, 2024.
[13] General Data Protection Regulation (GDPR) – Legal Text, gdpr-info.eu.
[14] Charter of Fundamental Rights of the European Union, European Parliament, December 18, 2000.
[15] Okechukwu, Amarachi Sharon. "Neuro-rights as a New Frontier of Human Rights: A Case of Recognition and Protection." *Alex-Ekwueme Federal University Faculty of Law LL. B Projects* (2024).
[16] Ibid.

secure the human right to participation in decision-making; and ensure free, prior, and informed consent.[17]

b. Second, **independent regulatory bodies should transition from reactive enforcement to proactive regulation** by evaluating "neurotechnology patents for potential social harms at the time of examination, and either directly notify relevant regulatory agencies or make evaluation results broadly available at the time of publication for granted patents."[18]
c. Finally, **technical safeguards such as encryption and privacy-preserving design should be mandated** to prevent unauthorized access or manipulation. These safeguards ensure that neurotechnology serves justice without eroding human dignity or fundamental freedoms.

6. Human Rights Protections

Human rights protections are facing technological, legal, and social threats that lead to discrimination and undermine equity and equality. Draft global standards on the ethics of neurotechnology underline that technologies must be developed and used in ways that avoid segregation, discrimination, or subordination of peoples, and promote gender-inclusive participation and representation throughout research and innovation.[19] The UNESCO draft calls for gender-responsive policies to close gaps in access and leadership in neurotechnology, highlighting the importance of inclusive governance in emerging scientific fields.[20] This reflects a broader shift in human rights approaches, in which participation, representation, and transparency are seen as essential safeguards against discrimination, rather than as optional policy goals.

Within the United States, the Department of Justice's Civil Rights Division has articulated a commitment to confronting discrimination in emerging technologies such as artificial intelligence, acknowledging the risk that AI systems may produce biased outcomes.[21] These concerns are especially relevant for racial and gender equity, since algorithmic systems can often reproduce historical biases present in data or institutional practices. Ethical frameworks such as the Belmont Report further emphasize justice and equitable treatment in research contexts by ensuring fair distribution of benefits and preventing exploitation of vulnerable groups.[22] Although developed for research on human subjects, its principles continue to guide contemporary discussions on fairness, accountability, and protection of vulnerable populations in technological development.

Taken together, these developments reflect evolving efforts to integrate gender equity and racial non-discrimination into human rights protections in law, policy, and technological governance. As innovation advances, maintaining human rights protections will depend on ensuring that technological systems are designed and governed to prevent discrimination and promote inclusion.

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[17] Ienca, Marcello. "On Neurorights." *Frontiers in Human Neuroscience* 15 (2021): 701258.
[18] Nadler, Roland, et al. "Harmful Uses of Patentable Neurotechnology: A New Regulatory Approach: Proposing a Framework Safeguarding Human Rights and Social Responsibility for Patented Applications of Neuroscience." *EMBO reports* 25, no. 5 (2024).
[19] MIT Media Lab, "Ethics of Neurotechnology: UNESCO Adopts the First Global Standard in the Cutting-Edge Technology," 2023.
[20] UNESCO, Draft Global Standard on the Ethics of Neurotechnology (2023) [referenced in MIT Media Lab article].
[21] U.S. Department of Justice, Civil Rights Division, "Artificial Intelligence and Civil Rights," https://www.justice.gov/archives/crt/ai.
[22] U.S. Department of Health and Human Services, "The Belmont Report," 1979.

7. Human Rights Impacts

Human Rights Impact Assessments (HRIAs) are an ex-ante human rights due-diligence tool. They assess affected rightsholders and foreseeable risks before a technology is designed and deployed, and they require those findings to be reflected in design choices, data governance, and operational standards.[23] In neurotechnology, this early timing matters because once neural data are collected, their use can readily expand through storage, analysis, inference, and reuse, and the resulting impacts on privacy, autonomy, and mental integrity are difficult to reverse. UNICEF cautions that in data-rich ecosystems, neurotechnology may threaten rights such as privacy and freedom of thought, and that wearable devices can enter the market without sufficiently rigorous oversight.[24] The 2019 controversy over EEG headbands in a Chinese primary school, which led local education authorities to suspend their use, suggests how neurotechnology can slip into everyday settings faster than safeguards when no independently verifiable prior assessment is in place.[25]

While the importance of HRIAs has gained attention, actual cases of corporate implementation and public disclosure remain limited. A representative case is *Girardi v. Emotiv Inc.*[26] Senator Guido Girardi Lavin purchased a device from U.S.-based company Emotiv and later found that the terms of service restricted non-subscribers' access to their neural data. The Chilean Supreme Court ruled that Emotiv had sold the device without adequate oversight of health and human rights, ordering the deletion of collected neural data and the suspension of sales until public health certification is obtained. This case showed that commercialization cannot be justified solely by click-through consent and emphasizes the need to institutionalize HRIAs as a pre-market requirement to protect privacy, mental integrity, and cognitive freedom.

8. Impacts on Marginalized Communities

As neurotechnology enters healthcare and legal fields, it exacerbates existing inequalities for marginalized communities, including for people of African descent and members of the LGBTQI+ community, through surveillance, biases, and weak accountability.

The human rights risks presented by neurotechnology can fall disproportionately on people of African descent and other marginalized communities who are underrepresented in neurotechnology datasets and decision-making power structures.[27] The training data for these algorithms is often not diverse and draws on historical records, making outputs less accurate and potentially harmful to people of African descent and other marginalized communities.[28]

There are also growing concerns about how neurotechnology may be used by institutions and governments. Since neurotechnology can affect cognition and emotion, its use in policing could expand into brain-reading of suspects or witnesses.[29] Error-prone identification and biased

[23] Danish Institute for Human Rights, *Welcome and Introduction: Human Rights Impact Assessment Guidance and Toolbox* (Copenhagen: Danish Institute for Human Rights, 2020).
[24] Eleonore Pauwels, *Neurotechnology and Children* (Florence: UNICEF Innocenti – Global Office of Research and Foresight, June 2024).
[25] Standaert, Michael. "Chinese Primary School Halts Trial of Device That Monitors Pupils' Brainwaves." *The Guardian.* November 1, 2019.
[26] *Girardi v. Emotiv Inc.,* Rol 105065-2023 (Corte Suprema de Chile, August 9, 2023).
[27] Eke, Damian. "Ethics and Governance of Neurotechnology in Africa: Lessons From AI." *JMIR Neurotechnology*, vol. 3, 2024, e56665.
[28] Murray, De-Shaine. "Neuroscience Has to Grapple With a Long Legacy of Racism if It Wants to Move Into the Future." *STAT*, 29 Nov. 2023.
[29] Bublitz, Christoph. "Neurotechnologies and Human Rights: Restating and Reaffirming the Multi-Layered Protection of the Person." *The International Journal of Human Rights*, vol. 28, no. 5, 2024.

prediction tools can also increase surveillance, arrests, and charging decisions that disproportionately affect people of African descent and other marginalized communities.

Neurotechnologies, such as brain-computer interfaces (BCIs), neural decoding systems, and affect recognition tools, also pose disproportionate risks to marginalized communities, particularly LGBTQI+ individuals, by intensifying surveillance and threatening cognitive privacy. Emerging "mind-reading" tools, for example, may infer sexual orientation or gender identity based on neural responses to stimuli, raising the possibility of involuntary outing in workplaces, medical contexts, or legal systems.[30]

Existing research also shows that neurotechnology datasets often lack representative diversity, resulting in algorithmic biases that disproportionately harm marginalized populations.[31] When LGBTQI+ individuals are underrepresented or misrepresented in neural datasets, neurotechnologies may produce inaccurate or pathologizing interpretations of their mental states. This mirrors broader patterns of technological bias in health and digital surveillance systems affecting LGBTQI+ communities.[32] **Robust regulatory safeguards and redistribution of decision-making power over neurotechnologies are required to prevent new forms of cognitive surveillance and discrimination and the acceleration of structural inequalities.**

9. Accountability and Access to Remedies

As we enter a new era of rapidly advancing neurotechnology and related innovations, the mechanisms for ensuring accountability and providing remedies when these technologies are misused in civil or criminal justice settings remain profoundly underexamined.

Given this, the United Nations needs to develop new human rights standards that reflect the emerging and rapidly evolving nature of neurotechnology. Until such standards are established, the use of neurotechnology, particularly in legal responses to violations of neurorights, should be subject to careful monitoring.

In the United States, the lack of national legislation on the development of neurotechnology is resulting in individual states taking piecemeal actions to regulate neurotechnology.[33] States such as California and Colorado have revised their existing privacy laws and added neural data protections into their legislation. However, greater collaboration with stakeholders and examination of what neural data protection means to different stakeholders is needed to inform policies and expand laws to cover grey areas.[34]

To begin addressing this issue, permissions should be clearly articulated when individuals are granted access to neurotechnology. Although this does not guarantee that their rights will be fully protected, ensuring users know what permissions they are granting is vital. At the same time, these permissions should be limited in scope. Individuals should not be able to consent to overly broad uses of neurotechnology until clear and established neurorights are in place. Because much remains unknown about the capabilities and risks of neurotechnology, the extent to which it has already been misused is unclear.

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[30] Emily Holmes et al., "Equity and Representation in Neurotechnology Research," *Journal of Medical Internet Research* 27, no. 1 (2025): e72325.
[31] Daniel Villarreal, "How Can We Stop Mind-Reading Technology from Outing Queer People Against Their Will?" *LGBTQ Nation*, March 2024.
[32] Sarah Jackson et al., "Digital Health Equity and Marginalized Populations," *JMIR Cancer* 12, no. 1 (2026): e81745.
[33] Laura Y. Cabrera et al., "Neurotechnology Governance in the United States: Gaps and Opportunities," *Bioethics* 40, no. 2 (2026): 225–35.
[34] Stanford Law School, "What Are Neural Data? An Invitation to Flexible Regulatory Implementation," Stanford Law School, December 2, 2024.

10. Judicial Decisions

Algorithmic discrimination has become an increasing concern for workers seeking employment, particularly since the beginning of the ongoing COVID-19 pandemic. Artificial intelligence now plays a significant role in the recruitment process. While exact figures vary by survey, research shows that a substantial share of employers and human resources departments use AI-assisted tools in hiring. Concerns about bias are well documented. Studies demonstrate that AI systems can replicate or amplify historical discrimination because they are trained on past data. A widely reported example occurred in 2018 when Amazon discontinued an experimental AI recruiting tool after it was found to disadvantage resumes associated with women.[35] Since then, AI-driven features have become common across hiring platforms and applicant-tracking systems. Legal challenges are beginning to test accountability. In *Saas v. Major Lindsey & Africa, LLC*, a plaintiff alleged gender and age discrimination related to recruiting practices, though the case did not result in a major precedent.[36] More prominently, *Mobley v. Workday, Inc.* involves allegations that an AI-driven hiring platform discriminated against an African American applicant over 40 based on race, age, and disability.[37] As the case proceeds, the United States still lacks comprehensive federal legislation specifically regulating AI in employment, leaving key questions unresolved.

Documented neurotechnology misuse, threatening privacy, autonomy, and mental integrity, demands durable legal safeguards. The European Union foregrounds a human-rights-centered paradigm exemplified by the Artificial Intelligence Act and the General Data Protection Regulation.[38] This approach recognizes how laissez-faire oversight leaves fundamental rights vulnerable to exploitation. By embedding principles of human dignity and data protection within binding supranational law, the EU ensures enforceable protections across Member States, thereby reducing the risk of fragmented compliance. This regulatory approach is especially pertinent for the United States, where Silicon Valley accounts for 55 percent of the AI market.[39] At least eight U.S. states have enacted legislation to protect neurorights, addressing concerns about cognitive liberty and biometric data.[40] To leverage this, the United States must implement a federal framework grounded in human rights law using the EU model. Such a framework must enshrine cognitive liberty, with independent oversight agencies and risk assessment, and guarantee accessible remedies, to help ensure fundamental rights are upheld together with innovation.

[35] Jeffrey Dastin, "Amazon Scraps Secret AI Recruiting Tool That Showed Bias Against Women," *Reuters*, 2018
[36] *SaaS v. Major Lindsey & Africa, LLC*, No. 1:23-cv-02102 (D. Md. 2023).
[37] *Mobley v. Workday, Inc.*, No. 3:23-cv-00770 (N.D. Cal. 2023).
[38] European Parliament. "Briefing: Protection of Mental Privacy."
[39] David Crawford. "AI's Trillion-Dollar Opportunity," https://www.bain.com/insights/ais-trillion-dollar-opportunity-tech-report-2024/.
[40] Neurorights Foundation. "US: Advocacy," https://www.neurorightsfoundation.org/advocacy/united-states.